

MESSAGE AUTHENTICATION CODE



CHAPTER 11

Outlines

1. Introduction to Message Authentication and Message Authentication Code.
2. Security on Message Authentication Code.
3. The Design of Practical Message Authentication Codes.

Learning Objectives

In the end of this Chapter 11, students will be able:

1. To understand the **concept and importance** of message authentication code in cryptography.
2. To understand the **security concerns** on message authentication code.
3. To learn the **design** of **practical message authentication codes**.

INTRODUCTION TO

MESSAGE AUTHENTICATION

Message Authentication Requirements

- In the context of communication across insecure network channel, message authentication is essential, as an adversary could launch any of the following attack:
 - 1. Disclosure** (Confidentiality). The release of message contents to any person or process not possessing the appropriate cryptographic key.
 - 2. Traffic Analysis** (Confidentiality). The discovery of the pattern of traffic between parties such as the length and number of encrypted message transmitted through the channel.

Message Authentication Requirements

3. **Masquerade** (Integrity). An insertion of messages into the network from a fraudulent source, which includes unauthorized creation and fraudulent acknowledgements of messages by unauthorized party.
4. **Content Modification** (Integrity). Changes to the contents of a message, including insertion, deletion, transposition, and modification.
5. **Sequence Modification** (Integrity). Any modification to a sequence of messages between parties, including insertion, deletion, and reordering.

Message Authentication Requirements

- 6. Timing Modification** (Integrity). Delay or replay of messages.
- 7. Source Repudiation** (Non-repudiation). Denial of transmission of message by source.
- 8. Destination Repudiation** (Non-repudiation). Denial of receipt of message by destination.

Message Authentication Functions

- Any message authentication or digital signature mechanism has **two levels of functionality**.
 - 1. Lower Level.** There must be some sort of function that **produces an authenticator**: a value to be used to authenticate a message.
 - 2. Higher Level.** This level uses the **lower-level function as a primitive** in an authentication protocol that enables a receiver to verify the authenticity of a message.

Message Authentication Functions

- The types of functions that produce an authenticator:
 - 1. Hash Function.** A function that maps a message of any length into a fixed-length hash value which serves as the authenticator.
 - 2. Message Encryption.** The ciphertext of the entire message serves as its authenticator.
 - 3. Message Authentication Code (MAC).** A function of the message and a secret key that produces a fixed-length value that serves as the authenticator.

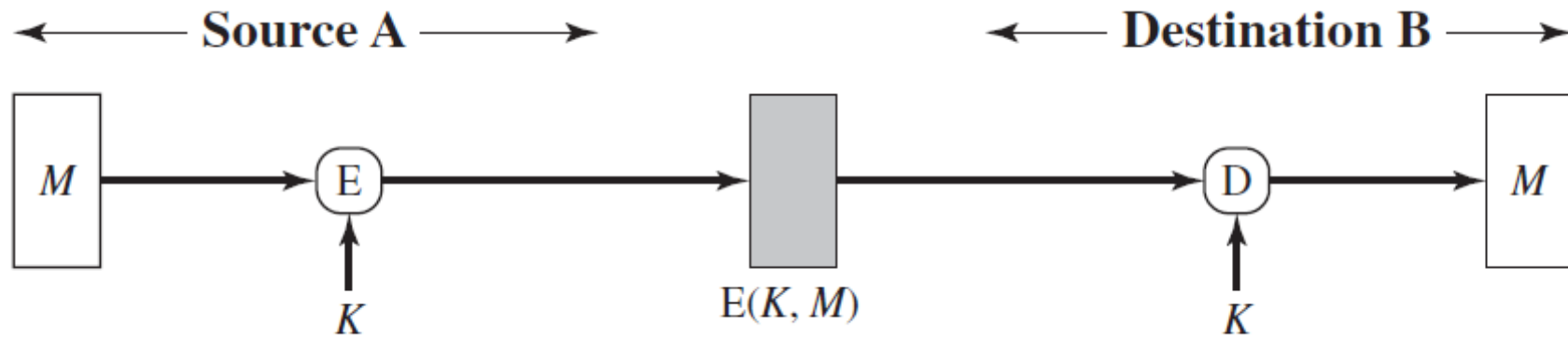
MESSAGE AUTHENTICATION FUNCTION

MESSAGE ENCRYPTION

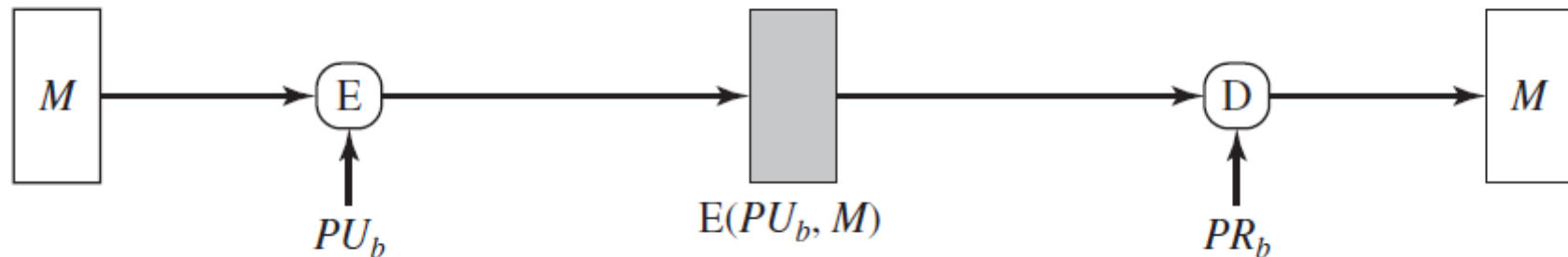
Message Authentication Functions - Message Encryption

- Message encryption by itself can provide a measure of authentication, either via symmetric encryption or public-key encryption.
1. **Symmetric** encryption that provide **confidentiality** and **authentication**.
 2. **Public-key** encryption that provide **confidentiality**.
 3. **Public-key** encryption that provide **authentication** and **signature**.
 4. **Public-key** encryption that provide **confidentiality**, **authentication** and **signature**.

Message Authentication Functions - Message Encryption

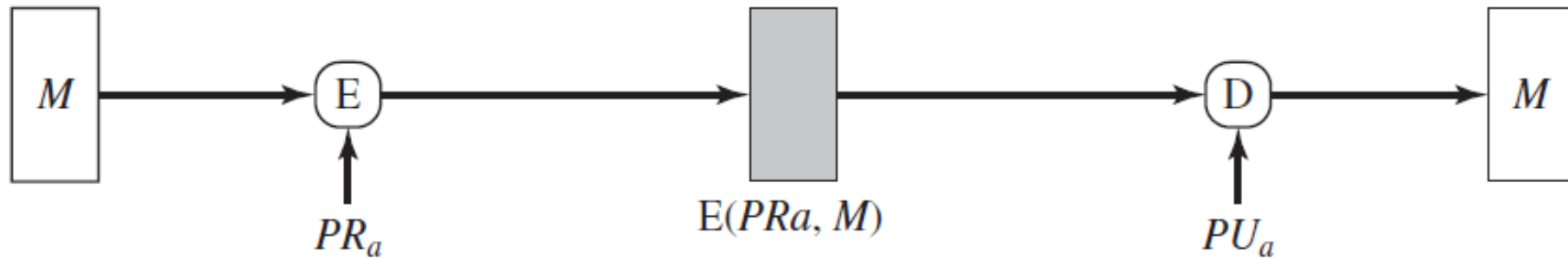


(a) Symmetric encryption: confidentiality and authentication

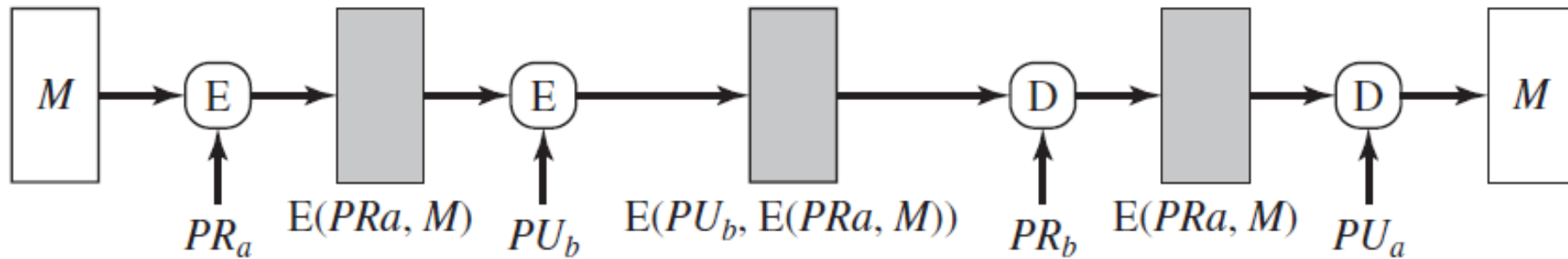


(b) Public-key encryption: confidentiality

Message Authentication Functions - Message Encryption



(c) Public-key encryption: authentication and signature



(d) Public-key encryption: confidentiality, authentication, and signature

MESSAGE AUTHENTICATION FUNCTION

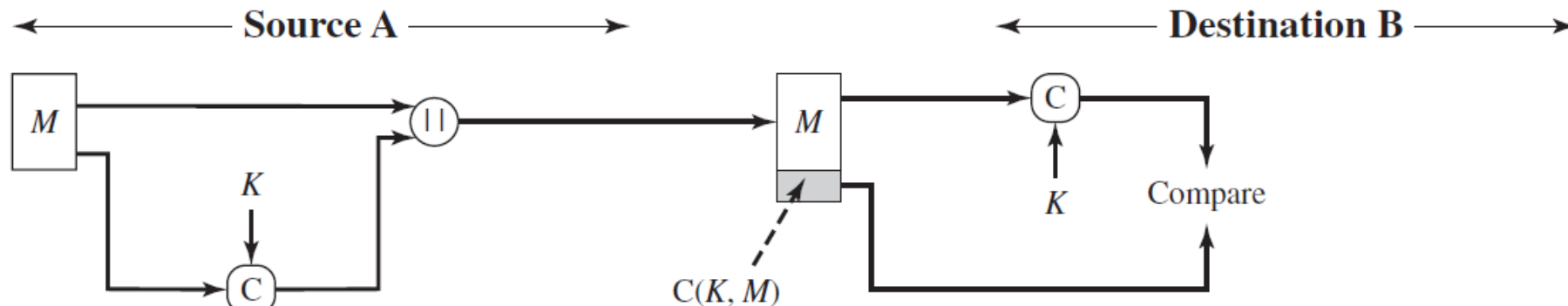
**MESSAGE AUTHENTICATION
CODE (MAC)**

Message Authentication Functions - Message Authentication Code (MAC)

- An alternative authentication technique involves the use of a secret key to generate a small fixed-size block of data, known as a **cryptographic checksum** or **MAC**, that is appended to the message.
- Suppose a sender and a receiver share a common secret key K .
 1. The sender wants to send a message M , he compute the MAC using the MAC function C and the secret key K , $MAC = C(K, M)$.
 2. The sender sends the concatenation of message M and the MAC as $(M \parallel MAC)$ to the receiver.

Message Authentication Functions - Message Authentication Code (MAC)

3. The receiver when receiving $(M \parallel \text{MAC})$, extracts M from it and computes $\text{MAC}' = C(K, M)$.
4. The receiver compares if $\text{MAC}' = \text{MAC}$. If it does, accept the MAC as valid, otherwise it rejects.

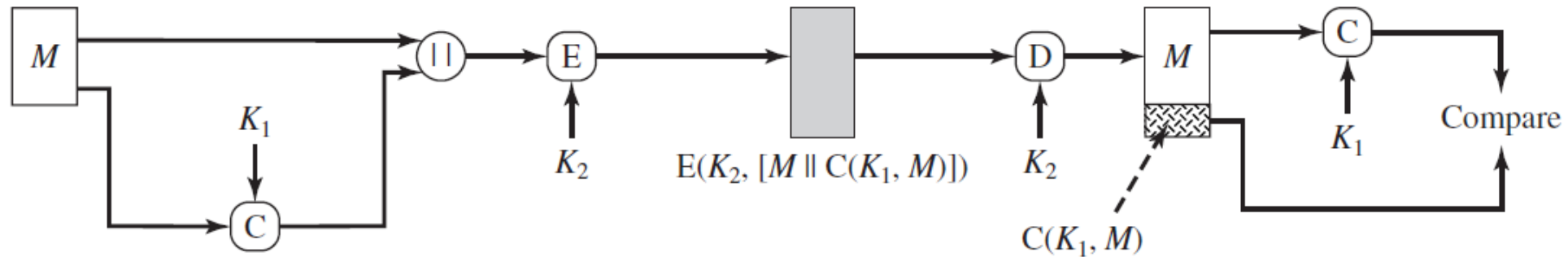


(a) Message authentication

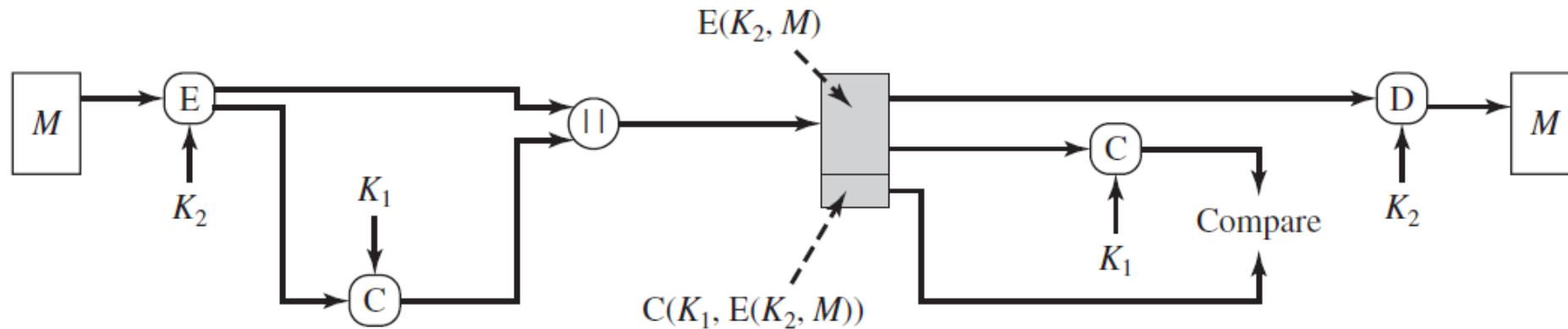
Message Authentication Functions - Message Authentication Code (MAC)

- A MAC function is similar to encryption with a difference that the MAC **algorithm need NOT be reversible**.
- In general, the MAC function is a **many-to-one function**.
- Besides the basic MAC, it can be extended to include confidentiality, with either:
 1. The authentication **tied to the plaintext**.
 2. The authentication **tied to the ciphertext**.

Message Authentication Functions - Message Authentication Code (MAC)



(b) Message authentication and confidentiality; authentication tied to plaintext



(c) Message authentication and confidentiality; authentication tied to ciphertext

Message Authentication Code (MAC) – Requirements

- Considering the security of the MAC, the design of MAC should satisfy the following requirements:
 1. Security against **Message Replacement Attacks**. A condition where an adversary is able to construct a new message to match a given MAC, even without knowing and learning the key used.
 2. Security against **Brute Force Attack**. This security is with respect to a chosen plaintext (message).
 3. Security on the **Authentication**. The authentication algorithm design should not be weaker with respect to certain parts of bits of the message.

MESSAGE AUTHENTICATION CODE

SECURITY

Attacks on MACs

- **Brute Force Attack.**

- This attack in MAC requires the **known message-tag pairs** $(x, H(x))$ to **find a collision** via picking a random bit string y and check whether $H(y) = H(x)$.

- There are ways brute force can be executed:

- 1. Attack the key space.** The attacker tries to determine the MAC key to generate a valid MAC value for any input of x .

- 2. Attack the MAC value.** The attacker tries to generate a valid tag for a given message, or to find a message that matches a given tag.

Cryptanalysis on MACs

- Cryptanalysis seek to exploit the property of an algorithm to perform specific attack other than exhaustive search.
- An ideal MAC Algorithm will require a cryptanalytic effort greater than or equal to the brute force effort.
- As there are various structures of MACs than in hash functions, discussing the general cryptanalysis is difficult here. Hence, we will NOT look into this section in MAC in this course.

DESIGN OF PRACTICAL MAC

HASH-BASED MAC (HMAC)

Hash-Based MAC – Motivations & Design

- In recent years, there has been increased interest in developing a MAC derived from a **cryptographic hash function**.
- **Motivations** behind such interest:
 1. Cryptographic hash functions such as MD5 and SHA generally **execute faster in software** than symmetric block ciphers such as DES.
 2. **Library code** for cryptographic hash functions is **widely available**.

Hash-Based MAC – Motivations & Design

- HMAC has been issued as **RFC 2104**, chosen as the mandatory-to-implement MAC for IP security, and is used in other Internet protocols, such as SSL. It has been issued as a NIST standard (FIPS 198).
- RFC 2104 lists the following **design objectives** for HMAC.
 1. To use, without modifications, available **hash functions**.
 2. To allow for **easy replaceability** of the embedded hash function in case faster or more secure hash functions are found or required.

Hash-Based MAC – Motivations & Design

3. To **preserve the original performance** of the hash function without incurring a significant degradation.
4. To use and handle keys in a **simple way**.
5. To have a well **understood cryptographic analysis of the strength** of the authentication mechanism based on reasonable assumptions about the embedded hash function.

Hash-Based MAC – Motivations & Design

H : Embedded hash function (e.g., MD5, SHA-1, RIPEMD-160).

IV : Initial Value input to hash function.

M : Message input to HMAC (including the padding specified in the embedded hash function).

Y_i : i th block of M , $0 \leq i \leq L - 1$.

L : Number of blocks in M .

b : Number of bits in a block.

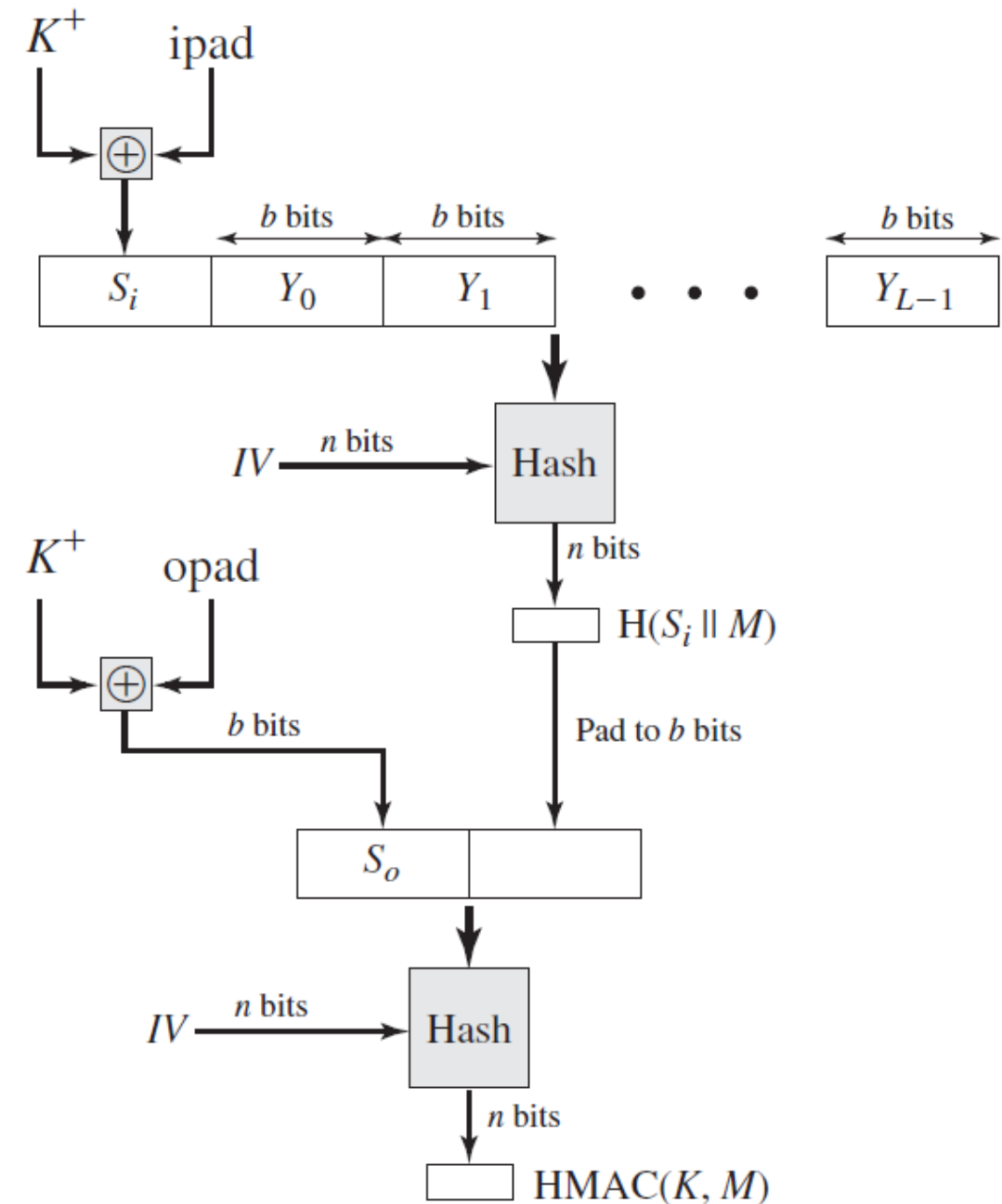


Figure 12.5 HMAC Structure

Hash-Based MAC – Motivations & Design

n : Length of hash code produced by embedded hash function.

K : Secret key (recommended length of at least n ; if key length is greater than b , the key is input to the hash function to produce an n -bit key.

K^+ : K padded with zeros on the left so that the result is b bits in length.

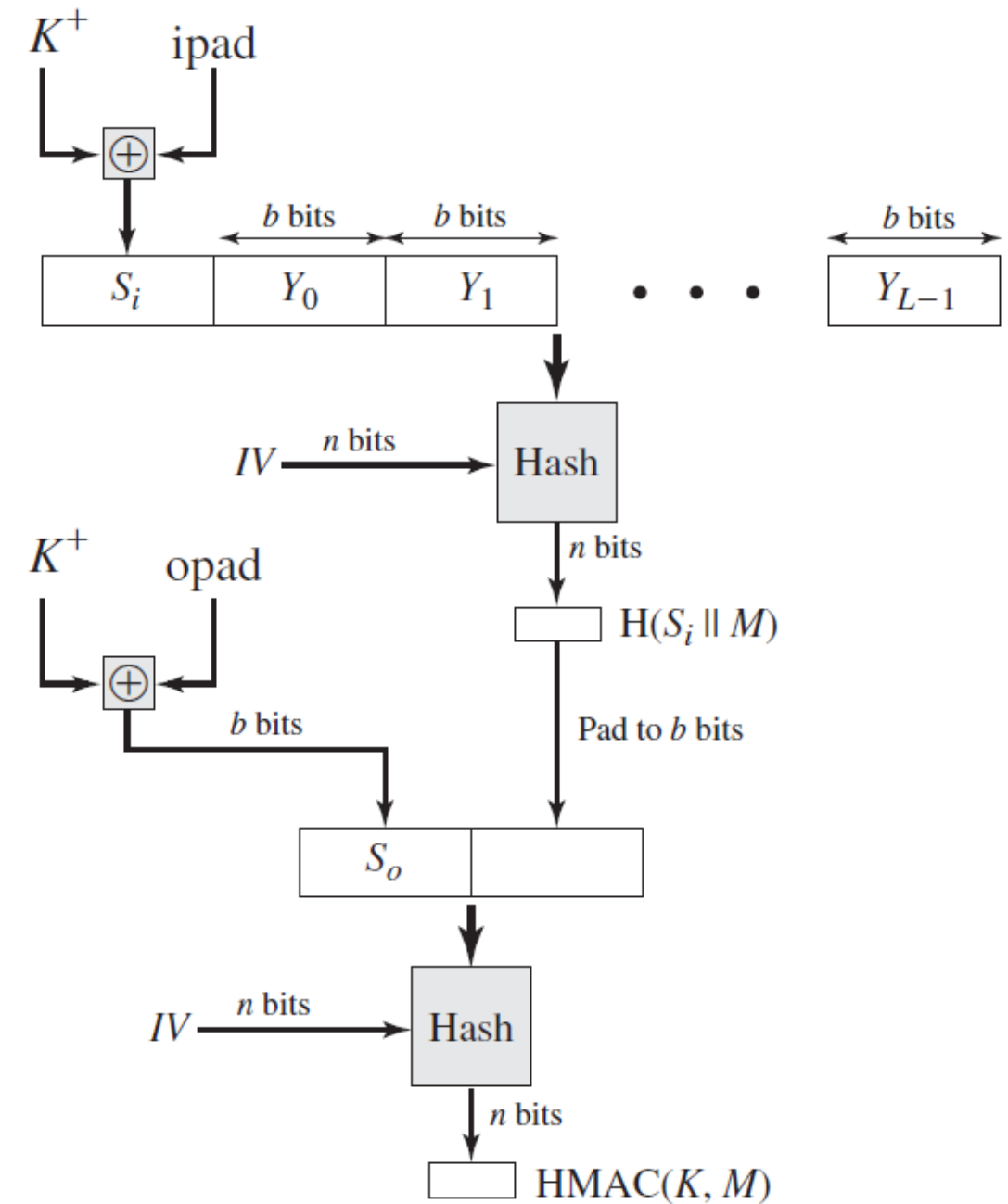


Figure 12.5 HMAC Structure

Hash-Based MAC – Motivations & Design

ipad: 0011 0110 (36_{16}) repeated $\frac{b}{8}$ times.

opad: 0101 1100 ($5C_{16}$) repeated $\frac{b}{8}$ times.

- Then HMAC can be expressed as:

$$\text{HMAC} = H[(K^+ \oplus \text{opad}) \parallel H[(K^+ \oplus \text{ipad}) \parallel M]]$$

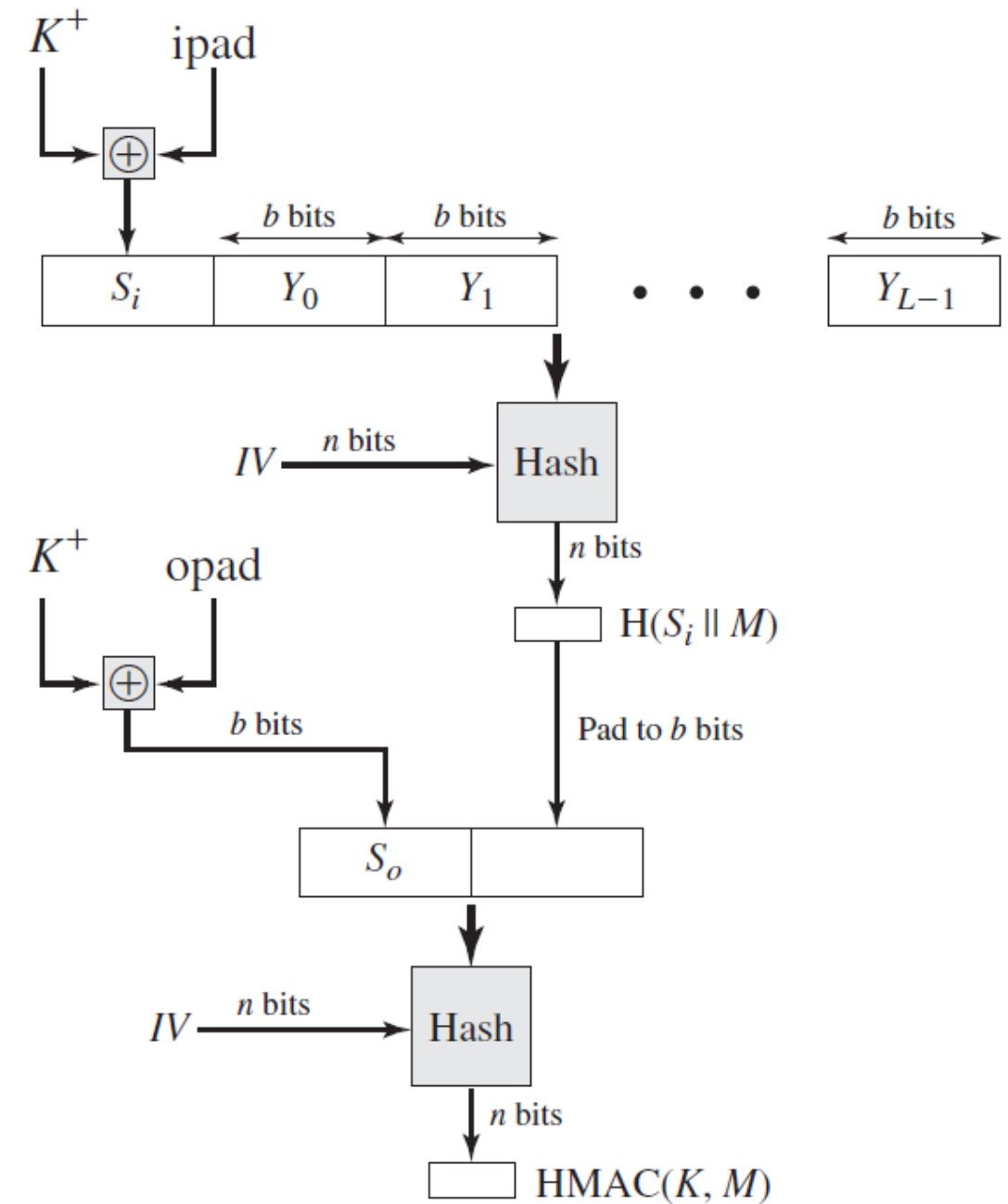


Figure 12.5 HMAC Structure

Hash-Based MAC – Security

- The security of HMAC **is equivalent** to on of the following attacks on the embedded hash function:
 1. The attacker is able to **compute an output** of the compression function even with an IV that is random, secret, and unknown to the attacker.
 2. The attacker **finds collisions** in the hash function even when the IV is random and secret.

DESIGN OF PRACTICAL MAC

MACs Based on BLOCK CIPHERS

MAC Based on Block Ciphers – Data Authentication Algorithm (DAA)

- Data Authentication Algorithm (DAA) uses **CBC mode** of **operation of DES** with an initialization vector of zero.
- The data to be authenticated are grouped in to contiguous **64-bit blocks**.
- A **Data Authentication Code (DAC)** is calculated using DES encryption algorithm and a secret key K .

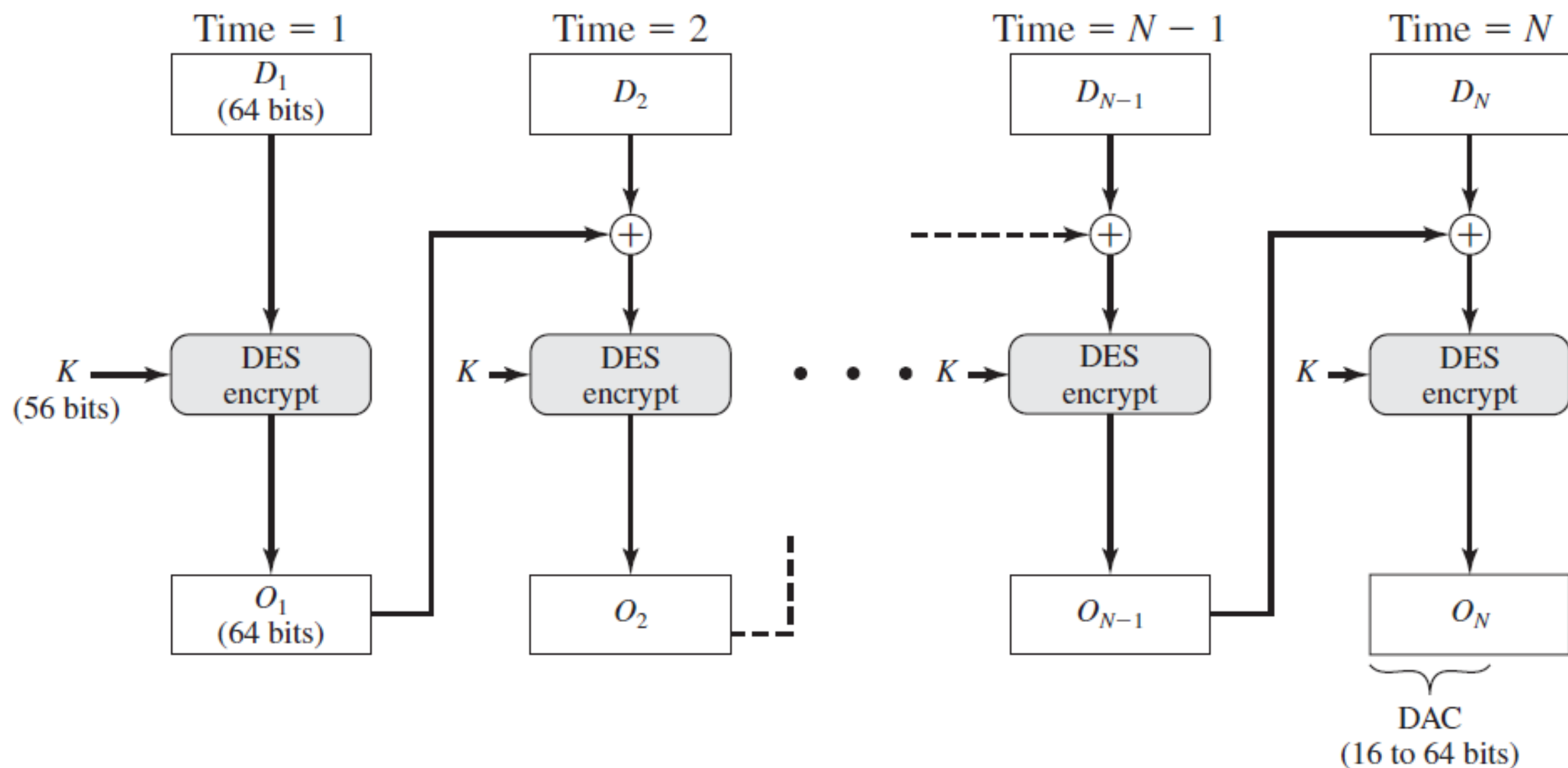
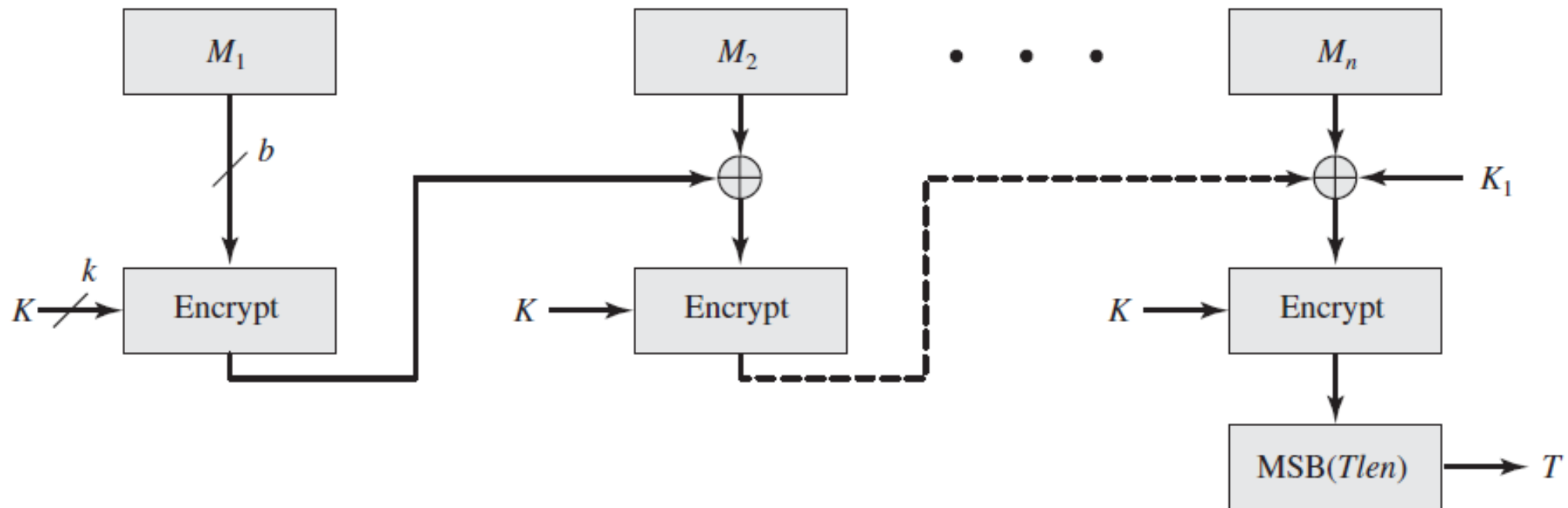


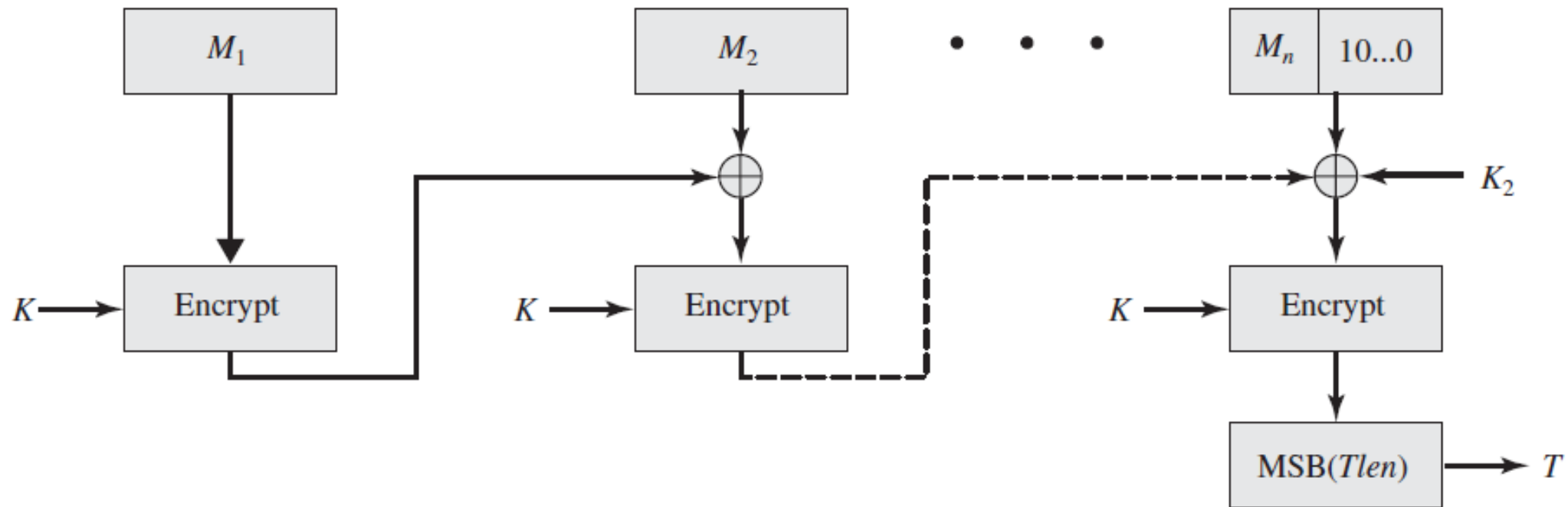
Figure 12.7 Data Authentication Algorithm (FIPS PUB 113)

MAC Based on Block Ciphers – Cipher-Based MAC (CMAC)



(a) Message length is integer multiple of block size

MAC Based on Block Ciphers – Cipher-Based MAC (CMAC)



(b) Message length is not integer multiple of block size

Message Authentication Code

OTHER
PRACTICAL MACs

Other Practical MACs

- **Authentication Encryption (AE)**. A term used to describe encryption systems that simultaneously protect **confidentiality** and **authenticity** of communications.
 1. Counter with Cipher Block Chaining-Message Authentication Code (CCM).
 2. Galois/Counter Mode (GCM).
- **Pseudorandom number generation (PRNG)**.

Main References for Chapter 11

1. Forouzan, B.A. "***Introduction to Cryptography and Network Security***", Third Edition, McGraw-Hill, 2008.
2. Stallings, W. "***Cryptography and Network Security: Principles and Practices***", Fifth Edition, Pearson Prentice Hall, 2011.

~ THE END 😊 ~